Research Statement

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I am a PhD student at Peking University, advised by Professor Xinyi Yuan, and expect to graduate in the summer of 2027. My research focuses on arithmetic geometry. I study heights and arithmetic intersection numbers on Shimura varieties, especially formulas that express them in terms of values and derivatives of L-functions. I am also interested in Arakelov geometry and use its methods throughout my work.

My completed work has three parts. Two papers compute Kodaira–Spencer maps on integral models and compare the metrics on the associated line bundles. A series of three papers computes modular heights of unitary Shimura varieties by comparing Eisenstein derivatives with arithmetic intersections of special divisors. A further paper computes heights of small CM cycles for general one-dimensional Hermitian embeddings. I am now working on the heights of more general CM points, intersections with unitary Shimura subvarieties, and holomorphic projection on $\mathrm{U}(2,2)$. Beyond these projects, I want to develop adelic height comparisons and explicit special value formulas on broader classes of Shimura varieties, using arithmetic intersection theory, automorphic forms, and relative trace formulas.

1 Completed work

I describe these three projects below. Links to the full papers are listed at the end.

1.1 Explicit Kodaira–Spencer maps

The Kodaira–Spencer map relates the invariant differentials of the universal abelian variety to differentials on the Shimura variety. Taking determinants compares the Hodge and canonical line bundles. I compute this comparison on integral models, including the metrics, for use in height formulas.

In [1], accepted by the *Kyoto Journal of Mathematics*, I treat Hilbert–Siegel and twisted Hilbert modular varieties. In [2], I extend the computation to PEL varieties of symplectic type and unitary varieties with equal positive and negative indices at every archimedean place. Let $\pi\colon \mathcal{A} \to \mathcal{X}$ be the universal abelian scheme on one of these integral models. Put $H = \det \pi_* \Omega^1_{\mathcal{A}/\mathcal{X}}$, and let $\omega_{\mathcal{X}}$ be the relative dualizing line.

Theorem 1.1. *Under the hypotheses and metric normalizations of [1, 2], there are fixed positive integers a, b, c determined by the PEL datum and integral lattice. The Kodaira–Spencer map induces a canonical injection*

$$\psi\colon H^{\otimes a} \hookrightarrow \omega_{\mathcal{X}}^{\otimes b}, \qquad \operatorname{im}\psi = c\,\omega_{\mathcal{X}}^{\otimes b}.$$

The map ψ is an isometry for the tensor powers of the Faltings and Petersson metrics.

Both papers compute a, b, c explicitly, including the discriminant contribution to c. For a Hilbert–Siegel modular variety of genus r, these constants are $a = r + 1$ and $b = c = 1$. Thus ψ gives an isometry

$$H^{\otimes(r+1)} \simeq \omega_{\mathcal{X}}.$$

In [2], I use Morita equivalence to compute determinants, then compute the integral image at ramified primes and compare the metrics exactly.

These results extend Yuan's calculation for quaternionic Shimura curves [8] and give explicit relations between the normalized arithmetic self-intersections of the Hodge and relative dualizing bundles. In [2, Section 1], I use these relations to compare existing formulas. On modular curves, they show that Yuan's formula for modular heights agrees with Kühn's arithmetic intersection formula [12, 9]. On quaternionic Shimura curves, they compare Yuan's formula with that of Kudla, Rapoport, and Yang [12, 20]. On unitary Shimura curves over imaginary quadratic fields, they compare my formula for modular heights with Bruinier–Howard's formula for arithmetic volumes [4, 7]. In this last comparison, the choice of line bundle gives a factor of two, while the different integral models contribute terms at ramified primes.

1.2 Modular heights of unitary Shimura varieties

In [3, 4, 5], I compute modular heights of unitary Shimura varieties in every positive dimension, under the field and local hypotheses of [4]. Let E/F be a CM extension with $F \neq \mathbb{Q}$. Let X have dimension $n \geq 1$, defined by a Hermitian space of signature $(n, 1)$ at one real place of F and positive definite at the others. Its tautological Hodge line L has a natural metric. Write $\overline{\mathcal{L}}$ for its integral extension on a certain integral model $\mathcal{X}$ fixed in [4]. The modular height is

$$h_{\overline{\mathcal{L}}}(X) = \frac{\widehat{\deg}(\overline{\mathcal{L}}^{n+1})}{\deg(L^n)}.$$

Write η for the quadratic Hecke character of E/F, L_f for the finite Hecke L-function, $d = [F : \mathbb{Q}]$, and $H_n = \sum_{j=1}^{n} 1/j$. Let d_F be the discriminant of F, $d_{E/F}$ the norm of the relative discriminant, and γ Euler's constant. I use the standard maximal lattices, which are self-dual at unramified places and modular or almost modular at ramified places according to the parity of n. I also use the Rapoport–Smithling–Zhang integral models, with the arithmetic normalization fixed in [4].

Theorem 1.2. *Under the hypotheses on the fields and integral models in [4], the modular height satisfies*

$$\begin{aligned} h_{\overline{\mathcal{L}}}(X) = 2\sum_{j=1}^{n+1} \frac{L_f'(j, \eta^j)}{L_f(j, \eta^j)} &+ 2(n+1)\log|d_F| \\ &+ \Big(n + 2 + \Big\lfloor \frac{n}{2} \Big\rfloor\Big) \log d_{E/F} \\ &+ d\big((n+1)H_n - n - 2(n+1)(\gamma + \log(2\pi))\big). \end{aligned}$$

Bruinier–Howard compute arithmetic volumes of unitary similitude varieties over imaginary quadratic fields [7]. My formula uses the tautological Hodge line over more general CM fields and different lattices at ramified primes. These choices require new local calculations and a comparison of generating series over a totally real field.

A special divisor on X is defined by an additional special homomorphism, whose norm gives its Fourier index. Its arithmetic class includes the divisor on the integral model and a Green function at infinity. An arithmetic Siegel–Weil formula relates arithmetic degrees or intersections of the generating series of these classes to derivatives of Eisenstein series. These identities are part of Kudla's program [19], which predicts that generating series of arithmetic special cycles should be modular. Pairing such arithmetic theta series with cusp forms gives arithmetic theta lifts. The program also predicts formulas relating the height pairings of these lifts to derivatives of L-functions.

Kudla, Rapoport, and Tonghai Yang prove such formulas on quaternionic Shimura curves [20]. For unitary varieties over imaginary quadratic fields, Bruinier, Howard, Kudla, Rapoport, and

Tonghai Yang prove modularity of arithmetic divisor series [21]. Bruinier, Howard, and Tonghai Yang relate pairings of arithmetic theta lifts with CM cycles to central L-derivatives [22].

My proof extends the method used in Yuan–Zhang–Zhang's Gross–Zagier formula, Yuan–Zhang's averaged Colmez formula, and Yuan's formula for modular heights of quaternionic Shimura curves [10, 13, 12]. Yuan–Zhang–Zhang compare the holomorphic projection of a mixed theta–Eisenstein derivative with a series of height pairings. Yuan–Zhang allow more general Schwartz functions to include CM self-intersections and recover CM heights. Yuan then considers a more general case and includes pairings with the Hodge class to compute the height of the whole curve. Pseudo-theta series in [13] and pseudo-Eisenstein series in [12] allow the constant term to be determined from the local comparisons over a totally real field.

The first paper [3] computes the holomorphic projection and local coefficients for the characteristic functions of the standard lattices. I use the projection methods of Yuan–Zhang–Zhang and Liu [10, 11], and adapt the comparison of pseudo-theta and pseudo-Eisenstein series to Hermitian spaces. The local computations give Whittaker derivatives at split, inert, and ramified places, together with archimedean kernels. These terms are compared with local intersections and Green functions.

The second paper [4] constructs the admissible arithmetic divisor series and computes its pairing with a compatible CM cycle, including divisors that contain the cycle. Comparing unitary and quaternionic Hodge bundles gives the initial curve case. The third paper [5] computes the arithmetic degrees of special divisors by intersecting them with powers of the Hodge bundle. I compare the restriction of the ambient Hodge bundle with the Hodge bundle on the lower-dimensional Shimura variety defined by a special divisor. Then I compute the difference of their integral structures and metrics, including the modular and almost modular cases at ramified primes. Comparing the constant terms of the analytic and arithmetic series then gives the modular height by induction on dimension.

These calculations also give an arithmetic Siegel–Weil comparison of rank one at positive Fourier indices, with explicit correction terms for the standard lattices. I use the CM pairings, divisor degrees, and comparisons of Hodge bundles in the projects described below.

1.3 Heights of small CM cycles

In [6], I compute the heights of small CM cycles on the unitary Shimura varieties above. Each cycle is defined by a one-dimensional Hermitian subspace $W \subset V$ which is negative at the distinguished real place and positive at the others. The earlier formula in [4] treats a particular compatible embedding. This paper allows general W relative to the fixed lattice Λ and determines how its integral position changes the height.

Let $\mathcal{P}_W$ be the normalized arithmetic CM cycle of degree one. At a finite place v of F, let $c_v \geq 0$ be the integer determined by $W_v \cap \Lambda_v$ as in [6, Section 2.1], and let N_v be the cardinality of the residue field. The integer c_v vanishes at almost all places.

Theorem 1.3. *Under the field and local hypotheses of [4], for any such W,*

$$\overline{\mathcal{L}} \cdot \mathcal{P}_W = -2\frac{L_f'(0,\eta)}{L_f(0,\eta)} + \sum_{v\nmid\infty} a_v(c_v)\log N_v,$$

where

$$a_v(c) = \begin{cases} c, & v \text{ split in } E/F, \\ c - \dfrac{2N_v}{N_v^2-1}(1-N_v^{-c}), & v \text{ inert in } E/F, \\ c - \dfrac{1-N_v^{-c}}{N_v-1}, & v \text{ ramified in } E/F. \end{cases}$$

The first term depends only on the CM extension E/F, while the local terms describe the embedding of W in the integral lattice. When every c_v is zero, the formula recovers the CM height in [4]. In [6], I also prove an arithmetic Siegel–Weil identity for these cycles, including explicit theta-series corrections arising at finite places. The proof compares local Whittaker derivatives with the lifting lengths of special homomorphisms and the restriction of arithmetic divisors to the CM cycle. The height formula also gives a Northcott finiteness result for small CM cycles with fixed ambient data. It holds for $F = \mathbb{Q}$ as well under the additional condition in [6, Theorem 1.1].

2 Future work

My immediate projects extend the divisor and CM-height calculations to general CM points, higher-dimensional Shimura subvarieties, and rank-two generating series. I also want to use adelic line bundles to study heights in families of Shimura subdata and, ultimately, develop exact arithmetic special value formulas beyond the unitary setting. The central problem throughout is to compare geometric and arithmetic invariants with automorphic L-functions while keeping track of the integral structures at finite places and the metrics at infinity.

2.1 Heights of general CM points

The small CM cycles of [6] arise from one-dimensional Hermitian subspaces over E. I now want to compute heights of CM points defined by more general CM algebras and CM tori. Fix a totally real extension F_1/F of degree $m \le n+1$ and put $K_1 = EF_1$. Suppose K_1 acts on an E-subspace $V_1 \subset V$, making V_1 one-dimensional over K_1 and containing the negative direction. Then

$$V = V_1 \perp V_+, \qquad \dim_E V_+ = n+1-m.$$

The norm-one torus of K_1/F_1 defines a CM cycle $\mathcal{P}$. More general CM data are described by totally real étale algebras whose other factors act on the positive part. In particular, this project allows CM fields of higher degree than those defining small CM cycles.

For $m < n+1$, I seek a formula of the shape

$$\overline{\mathcal{L}} \cdot \mathcal{P} = -\frac{2}{m}\frac{L_f'(0,\eta_1)}{L_f(0,\eta_1)} - \frac{1}{m}\log N_{F/\mathbb{Q}}\mathfrak{d}_{F_1/F} + \sum_{v\nmid\infty} b_v \log N_v,$$

where η_1 is the quadratic character of K_1/F_1, $L_f(s,\eta_1)$ is the finite Hecke L-function over F_1, $\mathfrak{d}_{F_1/F}$ is the relative discriminant, and the local coefficients b_v should be computed from the actual integral CM action and the position of the CM lattice. The case $m = 1$ is the formula already proved in [6].

My current calculations evaluate the finite Whittaker coefficients for maximal-order lattices and for general integral lattices through finite ideal-coset decompositions. The remaining work is arithmetic. I want to compare the actual polarized lifting functors and intersection ideals on relative Rapoport–Zink spaces, especially at ramified places, and compute the conormal maps of divisors containing the CM cycle. When $m = n + 1$, the positive rational complement is zero. The proof then uses a pure Eisenstein series, whose constant term and archimedean Green contribution require a separate comparison. My goal is to complete these calculations for both maximal and nonmaximal integral CM structures, and thereby compute heights of all CM points on these unitary Shimura varieties.

2.2 Arithmetic Siegel–Weil formulas in varying dimensions

In a second project, I replace the CM cycle by a unitary Shimura subvariety. Write $V = W^{\perp} \oplus W$, with $\dim_E W = m$ and $\dim_E W^{\perp} = n$. The variety X_W has dimension $m-1$, and X has dimension $m+n-1$. I first work with compatible standard lattices and integral maps $j_W \colon \mathcal{X}_W \to \mathcal{X}$ for which $j_W^* \overline{\mathcal{L}} \simeq \overline{\mathcal{L}}_W$, where $\overline{\mathcal{L}}_W$ is the metrized Hodge line on $\mathcal{X}_W$.

Let $\widehat{\mathcal{Z}}_*(g, \phi)$ be the arithmetic special divisor series without its constant term, for the standard Schwartz function ϕ. Let $I(s, g, \phi)$ be the mixed theta–Eisenstein series for $W^{\perp}$ and an incoherent Hermitian space of dimension m. Write $\deg_L X_W$ and $\deg_L X$ for the geometric degrees for the corresponding Hodge lines. The operator Pr is cuspidal holomorphic projection, and the derivative is taken at the Siegel–Weil point.

Target theorem 2.1. *Under the hypotheses on the fields and integral models in [4], let $m \geq 2$ and $n \geq 1$, with the compatible lattices and integral embeddings fixed above. There are explicit correction terms such that the following identity holds on* $\mathrm{U}(1,1)$*:*

$$\frac{\widehat{\deg}\big(j_W^* \widehat{\mathcal{Z}}_*(g,\phi) \cdot \overline{\mathcal{L}}_W^{m-1}\big)}{\deg_L X_W} - \frac{\widehat{\deg}\big(\widehat{\mathcal{Z}}_*(g,\phi) \cdot \overline{\mathcal{L}}^{m+n-1}\big)}{\deg_L X} = -2 \Pr I'(0, g, \phi) + \textit{explicit correction terms.}$$

The correction terms are Eisenstein and mixed theta–Eisenstein series determined by the local lattice data and the known modular heights.

Here the left side is the difference between a normalized intersection on X_W and a normalized arithmetic degree on X. My completed CM pairing for one-dimensional Hermitian subspaces [6] covers the case $m = 1$. This project replaces that pairing by intersections with positive-dimensional X_W, using the divisor degrees already computed on X. Divisors that contain X_W require arithmetic adjunction; the other divisors can be compared with divisors on X_W. I also need to compute the contributions of the Green functions at infinity. In fact, I want to compare every positive Fourier coefficient, then extend the formula to more general integral embeddings. Such an identity would explain how arithmetic intersections change under embeddings of unitary Shimura varieties.

2.3 Holomorphic projection on $\mathrm{U}(2,2)$

I am computing holomorphic projections of mixed theta–Eisenstein derivatives on $\mathrm{U}(2,2)$. The Fourier indices are now Hermitian matrices of size two. Both the Siegel and Klingen parabolics contribute boundary terms. Subtracting the corresponding Eisenstein corrections then gives a way to compute the projected Fourier coefficients. My current calculations treat a range of weights

where the integrals converge absolutely. Next, I want to treat low weights, especially weight three, where the Klingen contribution needs further work.

Then, I want to interpret these projected coefficients as arithmetic intersections and prove an arithmetic Siegel–Weil formula of rank two. A natural candidate is a generating series of arithmetic special cycles of codimension two, defined by pairs of special homomorphisms. The matrices of inner products of these pairs give the same Fourier indices. My plan is to determine which arithmetic degree or height pairing matches the mixed derivative, and compare the local intersections and Green functions with the projected coefficients. This would extend the comparison for divisors in my completed work to cycles of codimension two.

2.4 Adelic line bundles and canonical heights on Shimura varieties

I want to use Yuan–Zhang's theory of adelic line bundles [14] to reformulate the height comparisons of Pila–Shankar–Tsimerman [15]. In their proof of the André–Oort conjecture, the height of a special point is compared with a height attached to its Mumford–Tate torus. I am studying how to express this comparison through arithmetic degrees, using integral p-adic Hodge theory at finite places and CM periods at infinity.

Beyond this reformulation, I would like to study the precise arithmetic difference between the intrinsic metrics arising from p-adic Hodge theory and those induced by integral models of automorphic line bundles. This is particularly interesting at primes with non-hyperspecial level structures. As a first example, I plan to study an orthogonal Shimura variety associated with a quadratic lattice that is not self-dual at a finite place, and compute the resulting local height contributions at CM points. My longer-term goal is to obtain explicit comparisons that remain uniform as the CM data and level structures vary, refining the global height estimates of [15].

2.5 Special values on more general Shimura varieties

My longer-term goal is to relate heights and arithmetic intersections on Shimura varieties beyond unitary groups to special values and derivatives of L-functions. I would begin by identifying arithmetic cycles, automorphic line bundles, and the automorphic periods that should describe their heights. Depending on the setting, the analytic side may come from arithmetic Siegel–Weil formulas [23, 24], arithmetic theta lifts [25], or relative trace formulas [26]. The essential calculation is to match local intersections and Green functions with automorphic coefficients.

Orthogonal and GSpin Shimura varieties are natural first examples, since their special cycles and arithmetic volumes have been studied [16]. I would examine a Shimura curve arising from a suitable unitary subgroup inside an orthogonal Shimura threefold. Comparing the restriction of an automorphic line bundle with the intrinsic Hodge line bundle, and calculating the normal bundle and local intersections, should suggest a corresponding automorphic period. The relation between $\mathrm{SO}(3,2)$ and Siegel modular varieties of genus two may help identify the analytic side, while non-self-dual lattices require additional local calculations.

Furthermore, I am interested in exceptional Shimura varieties of Hermitian types $E_{6(-14)}$ and $E_{7(-25)}$. Recent constructions of integral models [17, 18] offer a starting point. For a family of CM points, I would choose an automorphic line bundle, identify its Mumford–Tate characters, and compare the Hodge metric with CM periods and the metrics from integral models. An initial goal is a height formula involving logarithmic derivatives of Artin or Hecke L-functions. I would also test these comparisons on classical Shimura subvarieties inside an exceptional variety, before discovering arithmetic special value formulas in more general settings.

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